Q3:

Question:

Can we create logical, similar groups of prospective customers in order to better understand the customer base? In addition, can we assign high likelihood conversions based off of these groups without needing to run a predictor on each segment?

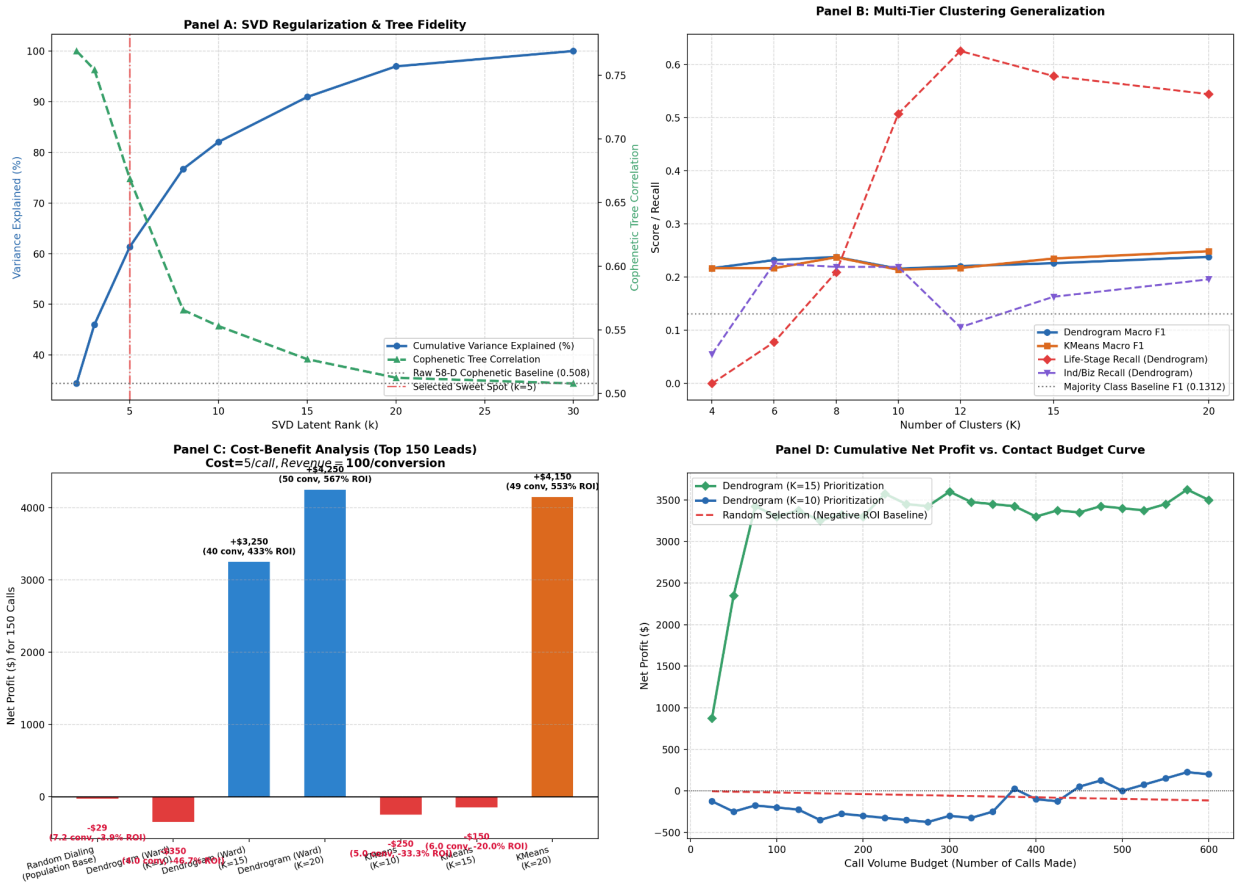
Approach:

The goal here was to create logical groups of clusters which have some relation, in order to create groups which we can prioritize. Since there were many features, this was a two step process: first, an SVD breakdown to reduce the number of features, and then a dendrogram / KMeans evaluation to see what ROI looked like with some assumptions as to cost of call and reward from a success. This was then evaluated using cross validation. We used a 5-fold cross-validation approach, with the preprocessing and SVD fitted independently within each training fold to avoid information leakage between training and validation data. Additionally, we withheld duration and y which are not collected before a call. We used a sweep to evaluate both KMeans and Dendrogram, and found a K=20 to be best for business performance for each, and an SVD latent rank of k=5 for dimensionality reduction. This gives a 567% ROI compared to randomly sampling prospective customers. The CBA was calculated using a $5 cost per call and a $100 reward per success, which are estimates of the economics of the campaign. This allowed us to further tune for the best CBA, since cost can make F1 an incomplete metric.

The SVD step is particularly important because it reduces the original 58-dimensional feature space into five latent components. These components capture combinations of the original features rather than simply selecting five individual columns. We evaluated different latent ranks to find a balance between retaining information and preserving the underlying hierarchical distance structure. At k=5, we retained approximately 61.4% of the original variance while achieving a cophenetic correlation of 0.669, compared to approximately 0.508 using the raw feature space. Importantly, the five latent dimensions do not limit us to five clusters: the subsequent hierarchical clustering can divide observations into 20 groups within this five-dimensional representation.

For the clustering evaluation, we tested different values of K (4, 6, 8, 10, 12, 15, and 20). Ward-linkage hierarchical clustering and KMeans were both evaluated. Since hierarchical clustering does not naturally provide a prediction method for unseen observations, we assigned validation observations to their nearest training-cluster centroid. KMeans used its standard prediction method. We then evaluated how well the resulting clusters generalized to different customer characteristics, including job tier, education, and housing status.

Evidence / Results:



SVD of k=5 reduced dimensionality from 58 to 5, which increased Cophenetic Fidelity from approximately 0.508 to 0.669. This was also included in a sweep which found it to be a reasonable balance between variance retention and tree fidelity. At this rank, approximately 61.4% of the original variance was retained. Higher ranks retained more variance, but progressively reduced cophenetic correlation toward the raw-data baseline. This suggests that dimensionality reduction helped remove noise and redundancy that could interfere with the hierarchical distance structure.

We ended up finding 4 distinct tiers of people, being: White Collar, Blue Collar/Services, Independent/Biz (including entrepreneurs and self-employed individuals), and "Life Stage" (Students/Retirees/Unemployed). The original job categories were consolidated into these broader groups to make the resulting segments easier to interpret. These tiers were then used to evaluate and explain the clustering results. Rather than treating the clusters as inherently meaningful labels, we examined which

job tiers were disproportionately represented within each cluster relative to their overall population prevalence. This allowed us to associate clusters with the customer characteristics they represented.

The clustering results showed that K=20 produced the strongest business outcomes for both methods. For the Dendrogram (Ward) approach, the top 150 prioritized calls produced 50 conversions, compared to approximately 7.2 expected conversions from random selection. This resulted in $4,250 in net profit and a 566.7% ROI. KMeans with K=20 produced 49 conversions and approximately 4,150 in net profit, with a 533% ROI. Both substantially outperformed random selection.

This demonstrates why optimizing purely for Macro F1 can be incomplete for this problem. A clustering configuration does not necessarily need to maximize overall classification performance to be economically useful. Instead, it needs to identify groups with sufficiently high conversion propensity to justify the cost of contacting them. Feeding our results into a CBA, we looked at calling high-propensity groups versus random dialing. This gave us an outcome of approximately +567% ROI for the Dendrogram approach, by prioritizing customers belonging to the highest-conversion-propensity clusters.

Overall, the results suggest that combining dimensionality reduction with hierarchical clustering can produce useful customer segments for targeted marketing. SVD k=5 provides a more compact representation of the data, while Dendrogram K=20 provides sufficiently granular segmentation to distinguish groups with substantially different conversion rates. The combination is therefore useful not only for describing customer populations, but also for making economically motivated decisions about which prospective customers should receive outreach.

Limitations:

SVD and Clustering only has so much fidelity, especially in a case like this with a rare positive class. This means we will likely not get maximum ROI since we are grouping individuals into groups rather than evaluating them individually, so we cannot discriminate between many high interest individuals. Additionally, since this model uses SVD and is not temporally conscious, if the customer base drifts over time, the model will likely take a very long time to adjust. SVD also necessarily loses some information, which would not be lost if we used a less dimensionally dependent model like a random forest.